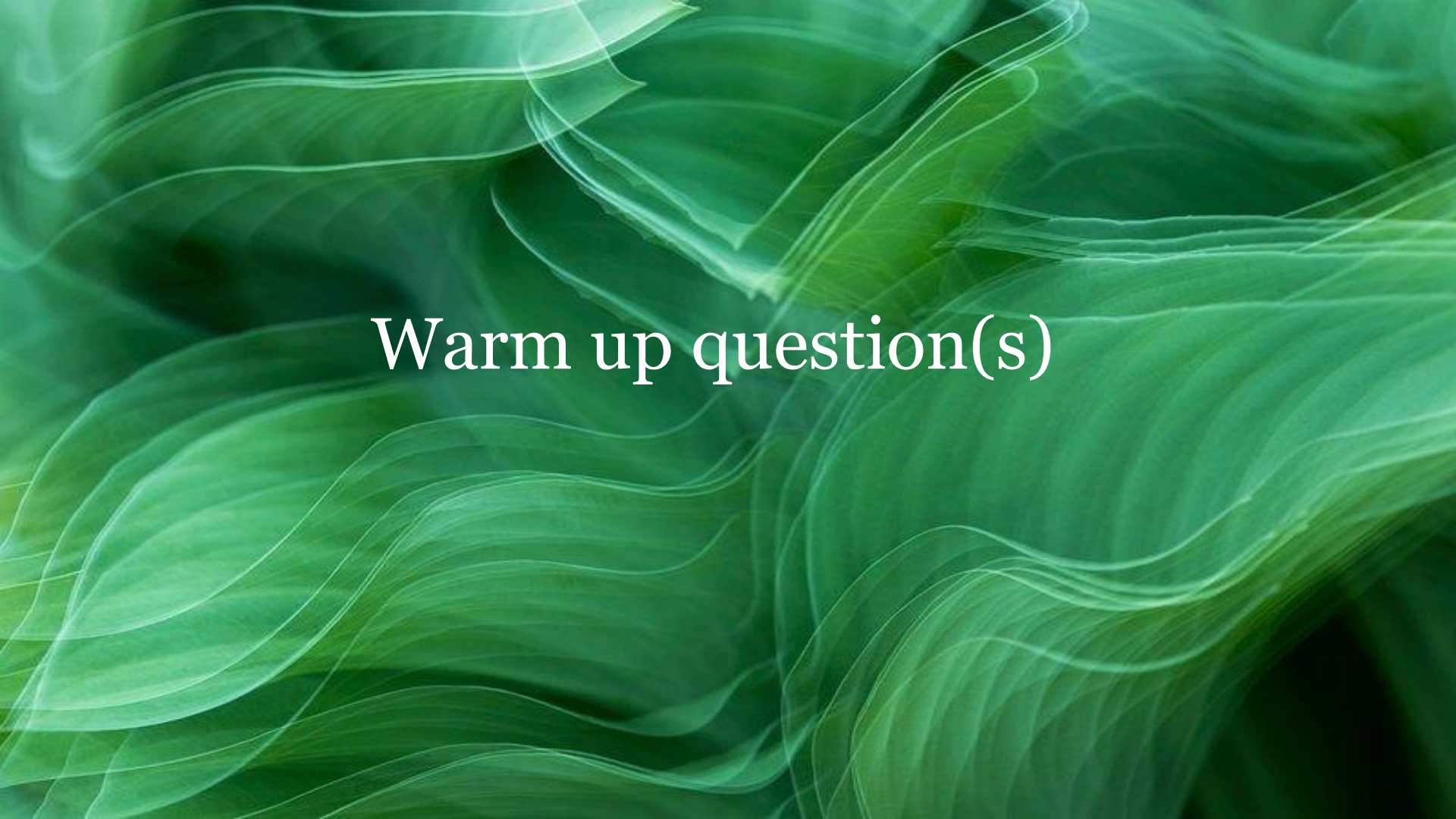


AI for Social Sciences

Dr Mary Sanford
Climate Change AI Summer School
22 July 2026



The background of the slide is an abstract, flowing pattern of various shades of green. The lines are wavy and layered, creating a sense of movement and depth. The colors range from a deep forest green to a lighter, almost lime green, with some areas appearing more translucent than others.

Warm up question(s)

Are you anxious about AI?



<https://partici.fi/97320810>

If so, why? If not, why not?



<https://partici.fi/97320810>

What are the social sciences?

Management/Finance

Government

Archaeology

Geography

Political science

Economics

Psychology

Education

Law

Sociology

International relations

Anthropology

Communications

My focus

Social science research on climate change

What do people think?

What do people do?

What influences people?

What about organisations?

What are the systemic implications?

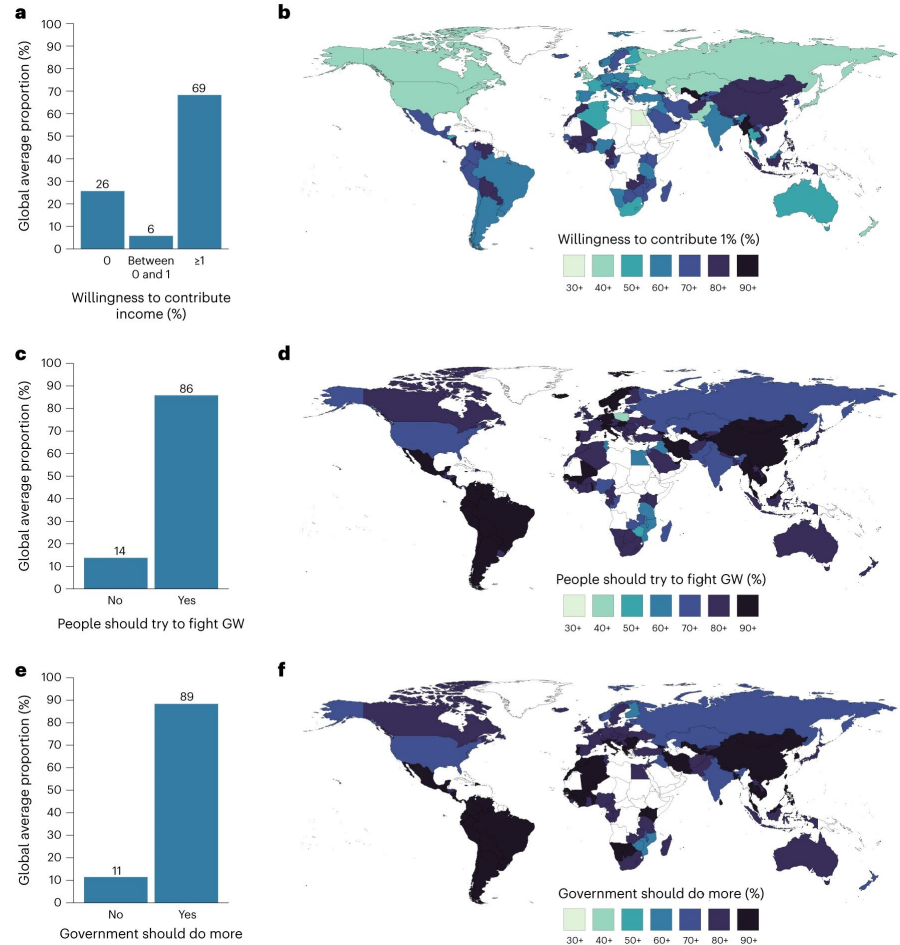
What can we learn from the past?

- Critical for building social and political feasibility for a just climate transition.
- **Necessary** for making sure technological advancements can reach their full potential.

What do people think?

- Big multinational surveys
- Self-reported support, opposition, attitudes, behaviours
- Mostly one-off, sometimes longitudinal
- A lot of descriptive work, some more causal/behavioural

Widespread global support for climate action





Journal of Environmental Psychology

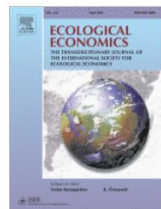
Volume 51, August 2017, Pages 226-238



“Green to be keep down”: effect of identity on pro-environmental behavior

Cameron Brick , David K. Sherman, Heejung S. Kim

- Some more experiments
- Secondary data analysis



Research article ● Full text access [Get rights and content](#) ➤

Do risk, time, and social preferences predict sustainable behavior? Evidence from a qualitative synthesis and meta-analysis ☆

Kristopher L. Nichols ^{a b} , Johanna M. Matt-Navarro ^a, Maxim K. Doiron ^a, Giovanna d'Adda ^{c d}, Elke U. Weber ^{a b e}, Sara M. Constantino ^{e f}

[Show more](#) ✓



The Political Consequences of Green Policies: Evidence from Italy

Published online by Cambridge University Press: 27 April 2023

[ITALO COLANTONE](#) , [LIVIO DI LONARDO](#) , [YOTAM MARGALIT](#) and [MARCO PERCOCO](#)

[Show author details](#) ✓



Global Environmental Change
Volume 17, Issues 3–4, August–October 2007, Pages 445–459



Barriers perceived to engaging with climate change among the UK public and their policy implications

Irene Lorenzoni ^{a b} , Sophie Nicholson-Cole ^b, Lorraine Whitmarsh ^b

What influences people?



What shapes public support for climate change mitigation policies? The role of descriptive social norms and elite cues

Published online by Cambridge University Press: 03 November 2020

[ADRIAN RINSCHIED](#) , [SILVIA PIANTA](#)  and [ELKE U. WEBER](#)

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European Journal of Political Research

Research Article |  Open Access |   

Carbon inequality and support for carbon taxation

[LIAM F. BEISER-MCGRATH](#)  [MARIUS R. BUSEMEYER](#)

First published: 15 December 2023 | <https://doi.org/10.1111/1475-6765.12647> | [VIEW METRICS](#)

“Fear Won’t Do It”: Promoting Positive Engagement With Climate Change Through Visual and Iconic Representations

[Saffron O'Neill](#) and [Sophie Nicholson-Cole](#) [View all authors and affiliations](#)

[Volume 30, Issue 3](#) | <https://doi.org/10.1177/1075547008329201>

Letter | Published: 30 November 2015

Network structure and influence of the climate change counter-movement

[Justin Farrell](#) 

[Nature Climate Change](#) **6**, 370–374 (2016) | [Cite this article](#)

Environment, Development and Sustainability (2025) 27:3593–3607
<https://doi.org/10.1007/s10668-023-04031-0>

How does personalized feedback on carbon emissions impact intended climate action?

[Yu Luo](#)¹  · [Brynley Hanson-Wright](#)² · [Hadi Dowlatabadi](#)² · [Jiaying Zhao](#)^{1,2}

Received: 26 April 2023 / Accepted: 9 October 2023 / Published online: 25 October 2023



What about organisations?

[Home](#) > [Crises and Disruptions in International Business](#) > Chapter

The Impact of Climate Risk on Firm Performance and Financing Choices: An International Comparison

Chapter | First Online: 26 April 2022

JOURNAL ARTICLE

Investor Rewards to Climate Responsibility: Stock-Price Responses to the Opposite Shocks of the 2016 and 2020 U.S. Elections

[Get access](#) >

[Stefano Ramelli](#), [Alexander F Wagner](#) ✉, [Richard J Zeckhauser](#), [Alexandre Ziegler](#)

The Review of Corporate Finance Studies, Volume 10, Issue 4, December 2021, Pages 748–787, <https://doi.org/10.1093/rcfs/cfab010>

Published: 17 June 2021 **Article history** ▼

Article | [Open access](#) | Published: 18 December 2023

Do fossil fuel firms reframe online climate and sustainability communication? A data-driven analysis

[Ramit Debnath](#) ✉, [Danny Ebanks](#), [Kamiar Mohaddes](#), [Thomas Roulet](#) & [R. Michael Alvarez](#)

npj Climate Action **2**, Article number: 47 (2023) | [Cite this article](#)

JOURNAL ARTICLE

Sectors, Pollution, and Trade: How Industrial Interests Shape Domestic Positions on Global Climate Agreements

[Federica Genovese](#)

International Studies Quarterly, Volume 63, Issue 4, December 2019, Pages 819–836,

<https://doi.org/10.1093/isq/sqz062>

Published: 27 August 2019

Systemic implications?

RESEARCH ARTICLE | SUSTAINABILITY SCIENCE | 



Social tipping dynamics for stabilizing Earth's climate by 2050

[Ilona M. Otto](#)  , [Jonathan F. Donges](#)  , [Roger Cremades](#)  ,  +10, and [Hans Joachim Schellnhuber](#)  [Authors Info](#)
[& Affiliations](#)

Contributed by Hans Joachim Schellnhuber, November 15, 2019 (sent for review January 22, 2019; reviewed by J. David Tabara and Jessika E. Trancik)

January 21, 2020 | 117 (5) 2354-2365 | <https://doi.org/10.1073/pnas.1900577117>



World Development

Volume 115, March 2019, Pages 246-257



Poverty and distributional effects of carbon pricing in low- and middle-income countries – A global comparative analysis

[Ira Irina Dorband](#)  , [Michael Jakob](#)  , [Matthias Kalkuhl](#)  , [Jan Christoph Steckel](#)  



Journal of Environmental Economics and
Management

Volume 127, September 2024, 103012



Climate change impacts on the within-country income distributions ☆

[Martino Gilli](#)  , [Matteo Calcaterra](#)  , [Johannes Emmerling](#)  ,
[Francesco Granella](#)  

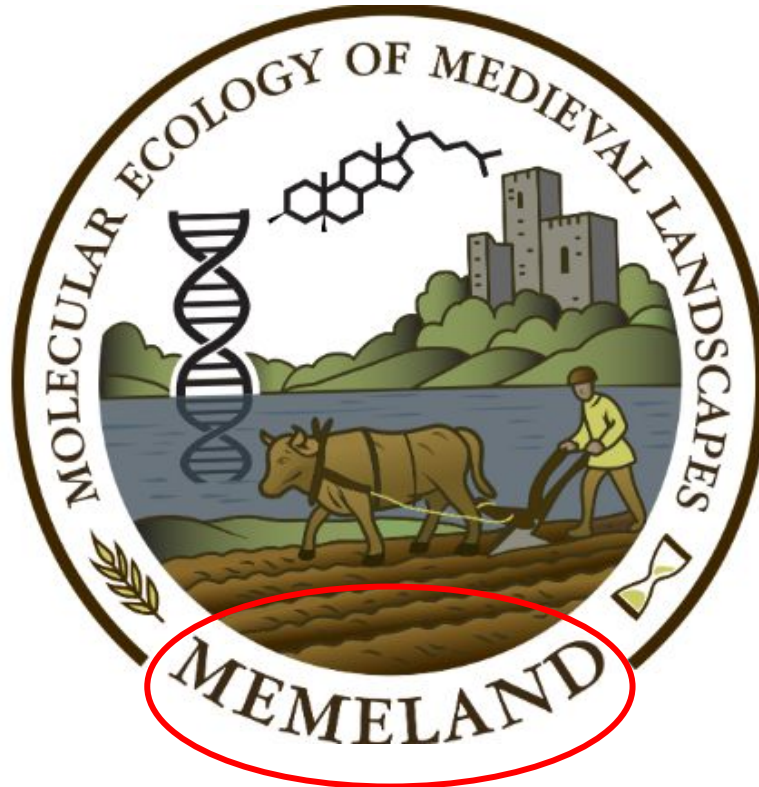
Analysis | Published: 29 November 2021

Climate action with revenue recycling has benefits for poverty, inequality and well-being

[Mark Budolfson](#) , [Francis Dennig](#), [Frank Errickson](#), [Simon Feindt](#), [Maddalena Ferranna](#), [Marc Fleurbaey](#),
[David Klenert](#), [Ulrike Kornek](#), [Kevin Kuruc](#), [Aurélie Méjean](#), [Wei Peng](#), [Noah Scovronick](#), [Dean Spears](#),
[Fabian Wagner](#) & [Stéphane Zuber](#)

Nature Climate Change 11, 1111–1116 (2021) | [Cite this article](#)

Learning from the past



But...not all social science research for
climate change is **quantitative**.

Qualitative work permits deeper understanding

- Critical for hypothesis development: identify and describe new phenomena
- Qualitative content analysis: investigate details and subjective angles that would get lost in quantitative analysis.
- Interviews, focus groups: ask questions to get further information that wouldn't be possible with surveys, observational analysis
- Can be paired with larger data collection efforts to test hypotheses at scale

Article | [Open access](#) | Published: 04 August 2025.

United Nations Environment Assembly attendees underestimate public willingness to contribute to climate action

[Ximeng Fang](#), [Joshua Ettinger](#) & [Stefania Innocenti](#) 

[Communications Earth & Environment](#) 6, Article number: 622 (2025) | [Cite this article](#)



Image matters: Climate change imagery in US, UK and Australian newspapers

Saffron J. O'Neill

Geography, College of Life & Environmental Sciences, University of Exeter, Amory Building, Rennes Drive, Exeter EX4 4RJ, UK



More than Symbols: The Effect of Symbolic Policies on Climate Policy Support

Published online by Cambridge University Press: 19 January 2026

[THEODORE TALLENT](#) , [MALO JAN](#)  and [LUIS SATTELMEYER](#)

[Show author details](#) ▾

ENVIRONMENTAL COMMUNICATION
2023, VOL. 17, NO. 7, 828–843
<https://doi.org/10.1080/17524032.2023.2259623>



 OPEN ACCESS  Check for updates

Extreme Weather Events as Teachable Moments: Catalyzing Climate Change Learning and Action Through Conversation

Joshua Ettinger ^a, Peter Walton ^b, James Painter ^c, Susan A. Flocke ^d and Friederike E.L. Otto ^e

Conceptual and theoretical work important too

- Developing and sharing new perspectives, ideas for research agendas, positions
- Building new theories
- Communicating within and across fields
- Very important for shaping research directions



Transportation Research Interdisciplinary
Perspectives

Volume 37, May 2026, 102029



Sufficient mobility and access within limits:
Research agenda for bringing together
corridor frameworks and transportation
research

Michał Czepkiewicz^{a,c}  , Giulio Mattioli^b  , Filip Schmidt^c  , Elias Willberg^d  ,
Lena Kilian^{e,n}  , Henrikki Tenkanen^f  , Dick Timmer^g  , Ákos Gosztonyi^h  ,
Johanna Raudseppⁱ  , Sanna Ala-Mantila^h  , Lisa Jacobson^j  , Mònica Guillen-Royo^k  ,
Dawid Krysiński^{l,c}  , Kevin Dillmanⁱ  , Jukka Heinonenⁱ  , Petter Næss^{m,1}

Perspective | Published: 30 September 2021

The role of high-socioeconomic-status people in locking in or rapidly reducing energy-driven greenhouse gas emissions

Dialogues on Climate Change

Volume 1, Issue 1, December 2024, Pages 23-29

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<https://doi.org/10.1177/29768659241293215>

Sage Journals

Commentaries



Are radical changes to lifestyles necessary for mitigating climate change?

Lorraine Whitmarsh¹ and Sam Hampton^{2,3}

Psychological Science in the Public Interest

Volume 23, Issue 2, October 2022, Pages 50-97

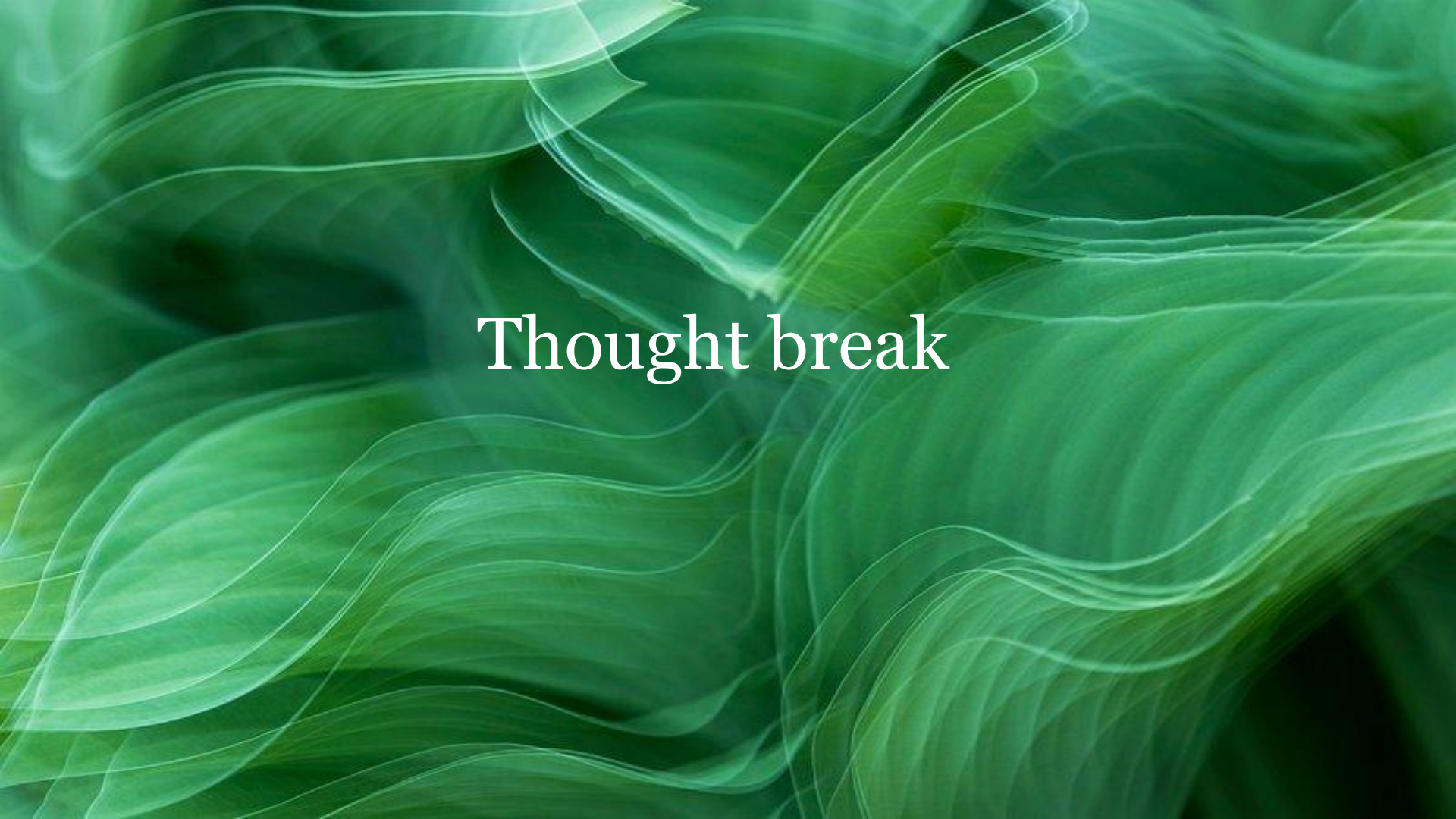
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<https://doi.org/10.1177/15291006221105279>

SAGE
journals

Scaling Up Change: A Critical Review and Practical Guide to Harnessing Social Norms for Climate Action

Sara M. Constantino^{1,2,3}, Gregg Sparkman⁴, Gordon T. Kraft-Todd⁴, Cristina Bicchieri^{5,6,7},
Damon Centola^{8,9}, Bettina Shell-Duncan¹⁰, Sonja Vogt^{11,12,13}, and Elke U. Weber^{3,14,15}

The background is an abstract composition of numerous overlapping, wavy, and flowing lines in various shades of green, ranging from light lime to deep forest green. The lines create a sense of movement and depth, resembling liquid or smoke. The text "Thought break" is centered in a white, serif font.

Thought break

Question

Can you think of a social science research topic or question related to climate change that you would like to learn more about?



<https://partici.fi/97320810>

Methodologically

Research process

How has AI changed the landscape?

Scope of research questions

Text classification at scale

The screenshot shows the Hugging Face website interface. The top navigation bar includes the Hugging Face logo, a search bar, and links for Models, Datasets, Spaces, Buckets, Docs, Enterprise, Pricing, Log In, and Sign Up. The left sidebar contains navigation links for Main, Tasks, Libraries, Languages, Licenses, and Other. The 'Models' tab is selected and highlighted with a red circle, showing 721 models. The 'climate' filter is applied. The main content area displays a grid of model cards for text classification tasks. Each card includes the model name, a brief description, and performance metrics. The models are sorted by trending.

Models 721 **climate** Base only Inference Available 11 Sort: Trending

Tasks

- Text Generation
- Any-to-Any
- Image-Text-to-Text
- Image-to-Text
- Image-to-Image
- Text-to-Image
- Text-to-Video
- Text-to-Speech
- +44

Parameters

< 1B 6B 12B 32B 128B > 500B

Libraries

- PyTorch TensorFlow JAX Transformers
- Diffusers GGUF MLX Transformers.js
- Safetensors sentence-transformers ONNX +42

Apps

- vLLM llama.cpp MLX LM Studio
- Ollama Jan Draw Things +9

Inference Providers

- Groq Novita Cerebras SambaNova
- Nscale fal Hyperbolic Together AI +11

Model Cards:

- Tino/Climate_Science_Stance**
Updated May 3, 2021
- cardiffnlp/bertweet-base-stance-climate**
Text Classification • Updated May 20, 2021 • ± 8 • $\heartsuit 1$
- cardiffnlp/twitter-roberta-base-stance-climate**
Text Classification • Updated Nov 28, 2022 • ± 61 • $\heartsuit 1$
- climatebert/distilroberta-base-climate-commitment**
Text Classification • $\pm 82.3M$ • Updated Jun 20, 2023 • ± 366 • $\heartsuit 2$
- climatebert/distilroberta-base-climate-d-s**
Fill-Mask • $\pm 82.4M$ • Updated May 4, 2023 • ± 7 • $\heartsuit 4$
- climatebert/distilroberta-base-climate-d**
Fill-Mask • Updated Apr 19, 2023 • ± 5 • $\heartsuit 2$
- climatebert/distilroberta-base-climate-detector**
Text Classification • $\pm 82.3M$ • Updated Jun 20, 2023 • $\pm 1.48k$ • $\heartsuit 17$
- climatebert/distilroberta-base-climate-f**
Fill-Mask • $\pm 82.4M$ • Updated May 4, 2023 • $\pm 4.87k$ • $\heartsuit 42$
- climatebert/distilroberta-base-climate-s**
Fill-Mask • $\pm 82.4M$ • Updated Jun 20, 2023 • ± 4 • $\heartsuit 3$
- climatebert/distilroberta-base-climate-sentiment**
Text Classification • $\pm 82.3M$ • Updated Jun 2, 2023 • $\pm 8.83k$ • $\heartsuit 5$
- climatebert/distilroberta-base-climate-specificity**
Text Classification • $\pm 82.3M$ • Updated Jun 20, 2023 • ± 784 • $\heartsuit 2$
- climatebert/distilroberta-base-climate-tcfd**
Text Classification • $\pm 82.3M$ • Updated Jun 20, 2023 • ± 188 • $\heartsuit 2$
- nlp-paradise/distilroberta-base-climate-f-finetuned..**
Updated Jan 24, 2022
- wiesnery/climate-roberta**
Updated Jan 23, 2022
- GPL/climate-fever-distilbert-tas-b-gpl-self_miner**
Sentence Similarity • Updated Mar 14, 2022 • ± 7
- Yoonseong/climatebert_trained**
Text Classification • Updated May 19, 2022 • ± 5 • $\heartsuit 1$
- amandakonet/climatebert-fact-checking**
Text Classification • Updated Feb 21, 2023 • ± 36 • $\heartsuit 10$
- GPL/climate-fever-msmarco-distilbert-gpl**
Sentence Similarity • Updated Apr 19, 2022 • ± 20
- GPL/climate-fever-tsdae-msmarco-distilbert-gpl**
Sentence Similarity • Updated Apr 19, 2022 • ± 17
- GPL/climate-fever-tsdae-msmarco-distilbert-margin-m..**
Feature Extraction • Updated Apr 19, 2022 • ± 12
- mwong/roberta-base-climate-evidence-related**
Text Classification • Updated Jun 24, 2022 • ± 6 • $\heartsuit 1$
- mwong/climatebert-base-f-fever-evidence-related**
Text Classification • Updated Jun 24, 2022 • ± 7 • $\heartsuit 2$

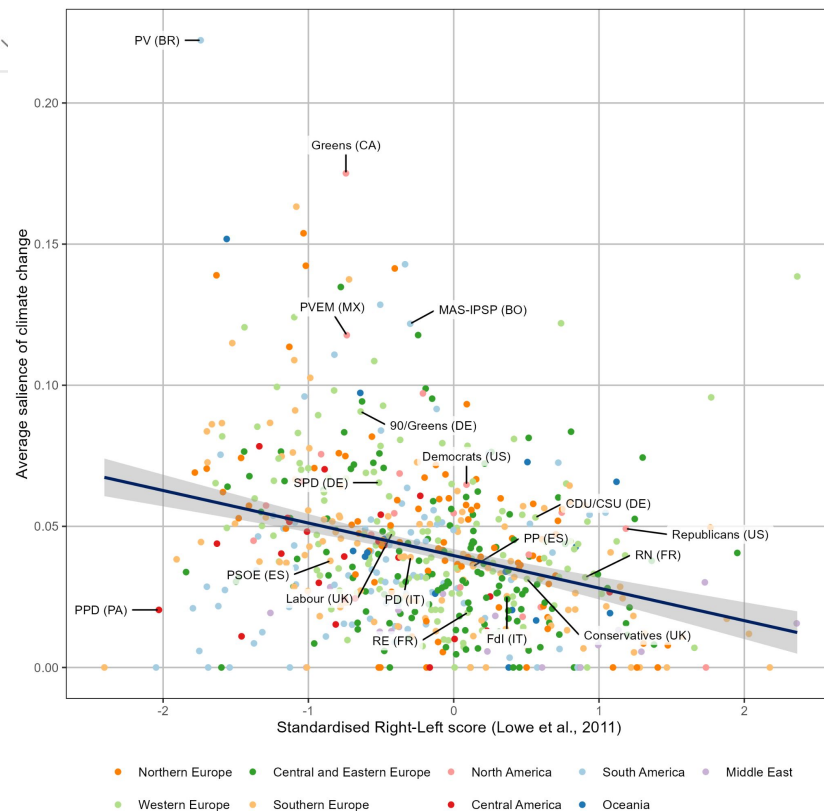
Policlim: A Dataset of Climate Change Discourse in the Political Manifestos of Forty-Five Countries from 1990 to 2022

Published online by Cambridge University Press: 23 October 2025

[Mary Sanford](#) , [Silvia Pianta](#) , [Nicolas Schmid](#)  and [Giorgio Musto](#) 

[Show author details](#) \

- Fine tunes **XLM-RoBERTa** to detect climate change salience in political manifestos across 20+ languages
- Model and variable ready for use in other work

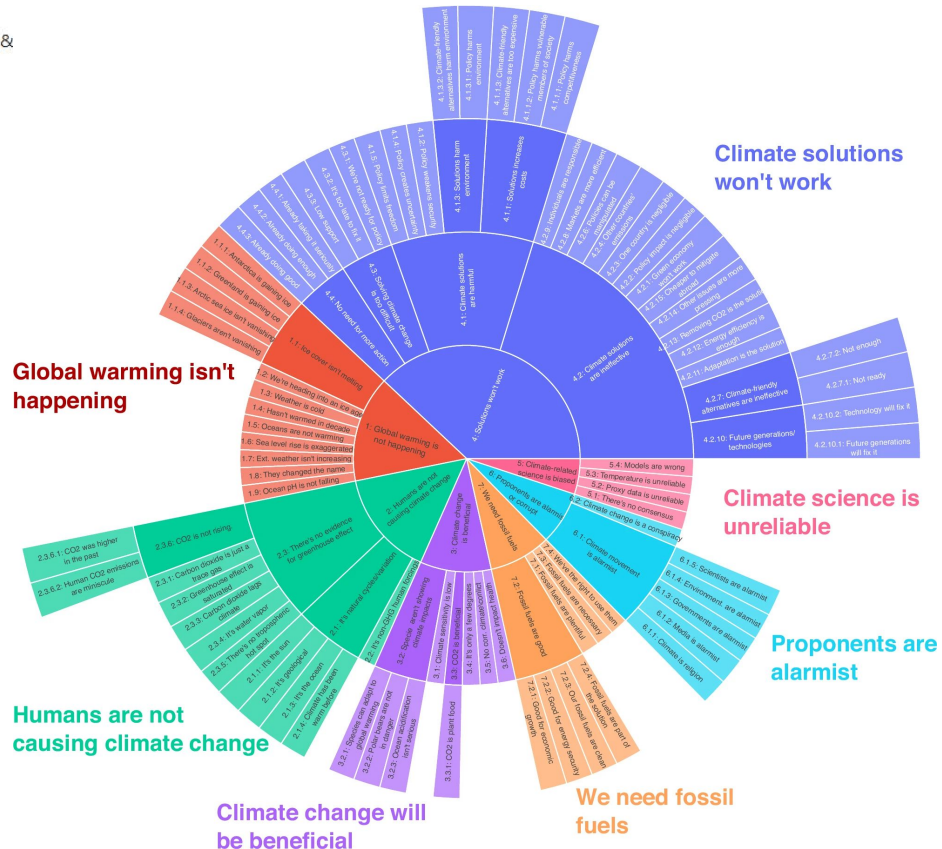


Large language model reveals an increase in climate contrarian speech in the United States Congress

[Travis G. Coan](#) , [Ranadheer Malla](#), [Mirjam O. Nanko](#), [William Kattrup](#), [J. Timmons Roberts](#), [John Cook](#) & [Constantine Boussalis](#)

[Communications Sustainability](#) 1, Article number: 37 (2026) | [Cite this article](#)

→ Takes complex taxonomy of contrarian claims



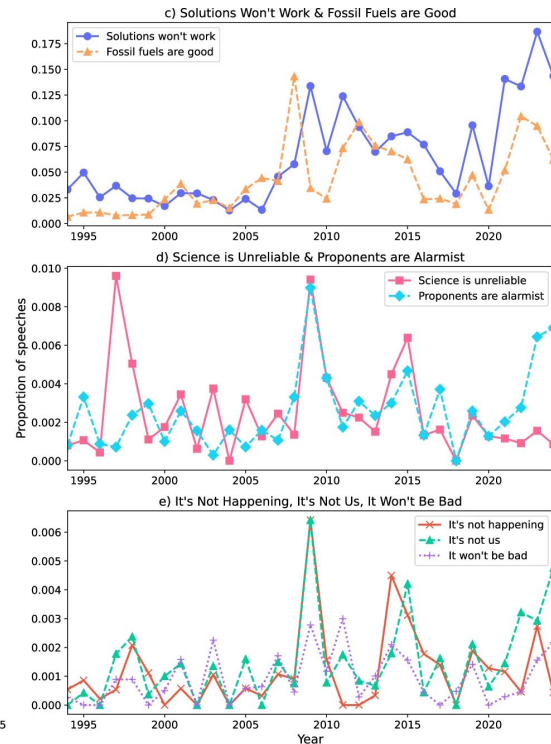
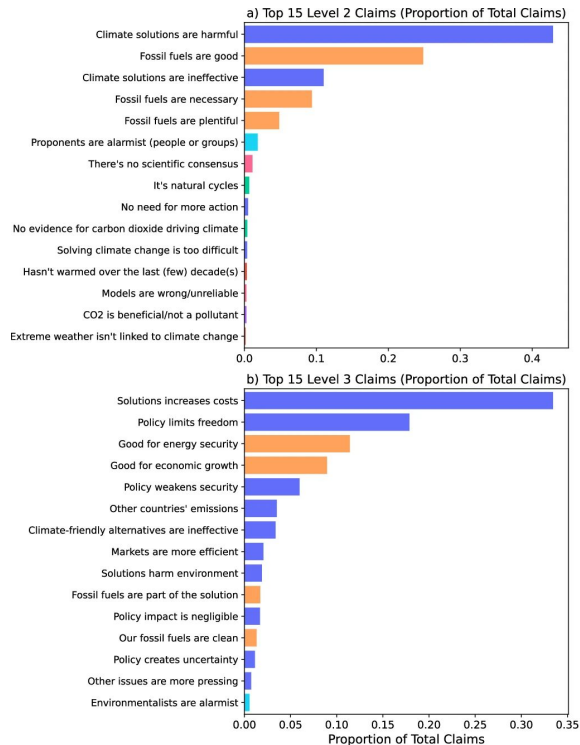
Large language model reveals an increase in climate contrarian speech in the United States Congress

[Travis G. Coan](#) , [Ranadheer Malla](#), [Mirjam O. Nanko](#), [William Kattrup](#), [J. Timmons Roberts](#), [John Cook](#) &

[Constantine Boussalis](#)

[Communications Sustainability](#) 1, Article number: 37 (2026) | [Cite this article](#)

- Takes complex taxonomy of contrarian claims and fine-tunes GPT4o and Claude Sonnet 3.7 to detect them.
- US Congressional speeches
- Track different narratives over time (and space)



Growing polarization around climate change on social media

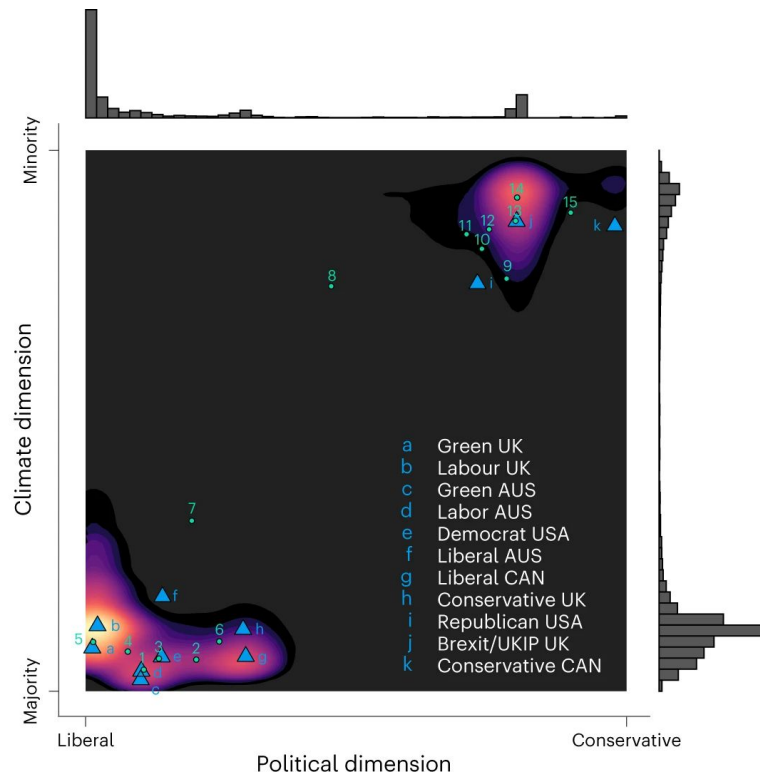
[Max Falkenberg](#), [Alessandro Galeazzi](#), [Maddalena Torricelli](#), [Niccolò Di Marco](#), [Francesca Larosa](#), [Madalina](#)

[Sas](#), [Amin Mekacher](#), [Warren Pearce](#), [Fabiana Zollo](#) ✉, [Walter Quattrociochi](#) ✉ & [Andrea Baronchelli](#) ✉

[Nature Climate Change](#) **12**, 1114–1121 (2022) | [Cite this article](#)

- Analyses retweet interactions between influential accounts in discourse around COPs to detect polarisation.
- Combines this with **BERTopic*** to further contextualise

- 1 Anti-Morrison
- 2 Biodiversity & Localism, Women at COP, Climate Finance, Youth, Climate Justice
- 3 Decolonise Climate
- 4 Sustainable Transport
- 5 Activists for Climate Action
- 6 Climate Crisis
- 7 Anti-Johnson
- 8 Anti-Sturgeon
- 9 Climate Alarmists
- 10 Wind Energy & Nuclear
- 11 Thunberg Cult
- 12 Chinese Absence, Vaccine Passports
- 13 Anti-Biden, Didn't Vote Green, Arctic Ice, Climate "Scam", Benefits of Warming
- 14 Illegal Immigration, Net Zero Referendum, Private Jets, Hypocrisy, CO₂
- 15 Anti-Trudeau



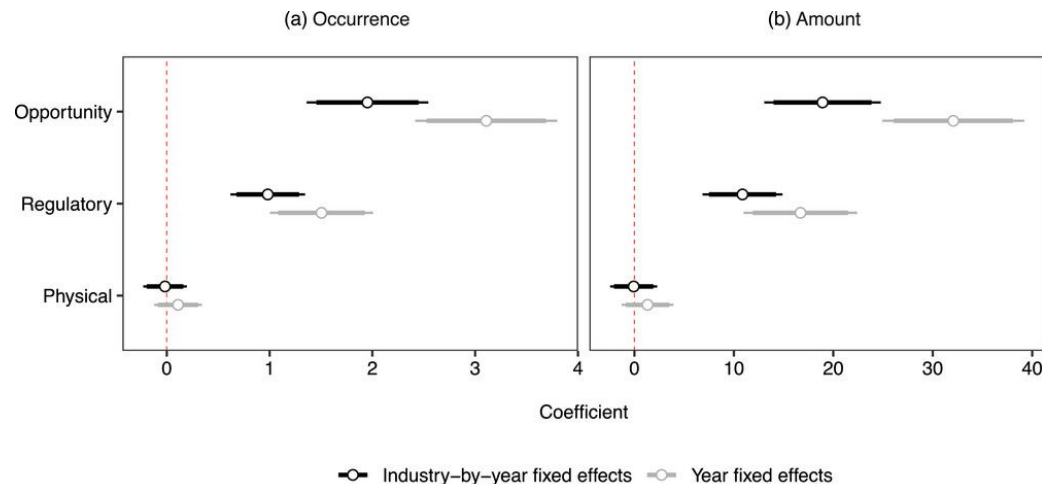
Climate exposure drives firm political behavior: Evidence from earnings calls and lobbying data

Christian Baehr, Fiona Bare, Vincent Heddesheimer ✉

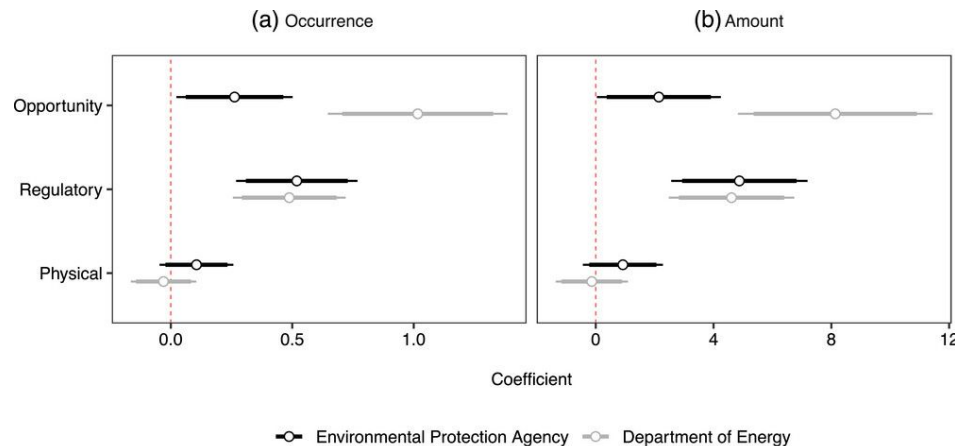
First published: 26 April 2026 | <https://doi.org/10.1111/ajps.70062> | [VIEW METRICS](#)

- Uses **ClimateBert** to analyse earnings calls
- Identify firms who mention opportunities related to climate change, as well as and regulatory and physical risks.
- Those who mention opportunities and regulatory risks more also tend to lobby more.

Effects of climate exposure on firm climate lobbying



Heterogeneous effects by lobbying target



Leveraging VLLMs for Visual Clustering: Image-to-Text Mapping Shows Increased Semantic Capabilities and Interpretability

[Luigi Arminio](#) , [Matteo Magnani](#)  , [...], and [Alexandra Segerberg](#)  [View all authors and affiliations](#)

[Volume 44, Issue 3](#) | <https://doi.org/10.1177/08944393251376703> | [View article versions](#)

- Use VLLMs (gpt-4-turbo and LLaVa) to generate text descriptions of images
- Then use this to cluster the images for further quantitative and qualitative analysis
- *“One aspect that should be checked is the **potential bias** introduced in the analysis, both with respect to **possible mistakes**, but also with respect to **possible preferences** assigned to one instead of a different cluster”*





Misclassification bias

- No prediction is perfect
- Far too few development pipelines contain thorough validation checks
- Results in bias in downstream analysis
- Even more problematic for LLMs

Large Language Model Hacking: Quantifying the Hidden Risks of Using LLMs for Text Annotation

Joachim Baumann¹, Paul Röttger¹, Aleksandra Urman², Albert Wendsjö³,
Flor Miriam Plaza-del-Arco⁴, Johannes B. Gruber⁵, and Dirk Hovy¹

¹Bocconi University

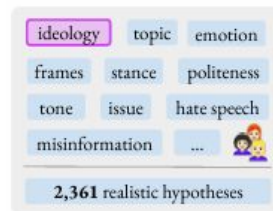
²University of Zurich

³University of Gothenburg

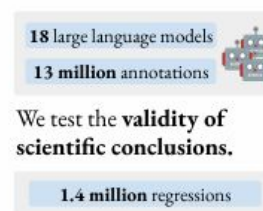
⁴LIACS, Leiden University

⁵GESIS, Leibniz Institute for the Social Sciences

We replicate 37 social science
data annotation tasks.



LLMs take over the role of
data annotators.



LLM annotations are unreliable
for downstream statistical analyses!

Task: Is manifesto *left* or *right*?
Ground truth: no effect ($p > 0.05$)

LLM 1, config 1	$p = 0.242$	✓
LLM 1, config 2	$p = 0.001$	✗
LLM 2, config 1	$p = 0.424$	✓
...	...	...

LLM hacking risk = 31% – 50%

Misclassification bias

Before using a new model...

- Always check the training data (examining examples if possible) to understand if the context is appropriate
- Take steps to mitigate bias where possible (TeBlunthuis et al., 2024)
- Avoid basing model performance evaluations only on training. Always validate predictions too.

COMMUNICATION METHODS AND MEASURES
2024, VOL. 18, NO. 3, 278–299
<https://doi.org/10.1080/19312458.2023.2293713>

 **Routledge**
Taylor & Francis Group



Misclassification in Automated Content Analysis Causes Bias in Regression. Can We Fix It? Yes We Can!

Nathan TeBlunthuis ^{a,b}, Valerie Hase ^c, and Chung-Hong Chan ^d

Survey analysis



Energy Research & Social Science

Volume 73, March 2021, 101883



Why hate carbon taxes? Machine learning evidence on the roles of personal responsibility, trust, revenue recycling, and other factors across 23 European countries

Sebastian Levi

Research article Open access Under a Creative Commons [license](#) [Get rights and content](#)

Unpacking the side effects of social norm nudges

[Julien Picard](#) ^{a b c} , [Sanchayan Banerjee](#) ^d [Show more](#)

- New ways of studying relationships between variables, e.g., discover latent structures, predictive strength.
- Can increase precision in models.
- Can boost [causal inference pathways](#) depending on data and context.
- Important to address uncertainty.
- *Always needs theoretical grounding.*

Policy evaluation

RESEARCH ARTICLE | CLIMATE POLICY



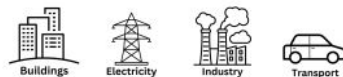
Climate policies that achieved major emission reductions: Global evidence from two decades

ANNIKA STECHEMESSER, NICOLAS KOCH, EBBA MARK, ELINA DILGER, PATRICK KLÖSEL, LAURA MENICACCI, DANIEL NACHTIGALL, FELIX PRETIS

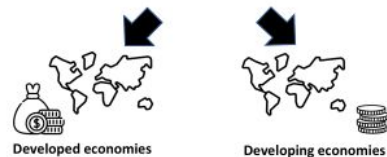
NOLAN RITTER, [...] AND ANNA WENZEL +2 authors [Authors Info & Affiliations](#)

SCIENCE • 22 Aug 2024 • Vol 385, Issue 6711 • pp. 884-892 • DOI: 10.1126/science.adl6547

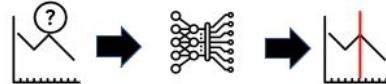
CO₂ Emissions data from 4 sectors (2000-2020)



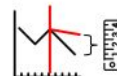
Split into two country groups



Agnostically detect breaks in emissions using Machine Learning for each sector and country group



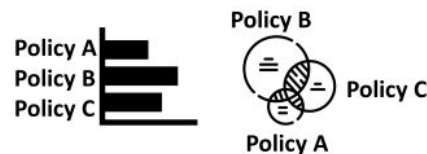
Estimate the treatment effect of each break with causal inference method (i.e. Difference-in-differences)



Attribute breaks to high-quality sectoral climate policy data



Analyse successful policies and policy mixes



Evidence synthesis

Article | Published: 11 October 2021

Machine-learning-based evidence and attribution mapping of 100,000 climate impact studies

[Max Callaghan](#) , [Carl-Friedrich Schleussner](#), [Shruti Nath](#), [Quentin Lejeune](#), [Thomas R. Knutson](#), [Markus Reichstein](#), [Gerrit Hansen](#), [Emily Theokritoff](#), [Marina Andrijevic](#), [Robert J. Brecha](#), [Michael Hegarty](#), [Chelsea Jones](#), [Kaylin Lee](#), [Agathe Lucas](#), [Nicole van Maanen](#), [Inga Menke](#), [Peter Pfleiderer](#), [Burcu Yesil](#) & [Jan C. Minx](#) 

[Nature Climate Change](#) **11**, 966–972 (2021) | [Cite this article](#)

Article | [Open access](#) | Published: 07 September 2024

Public support for carbon pricing policies and revenue recycling options: a systematic review and meta-analysis of the survey literature

[Farah Mohammadzadeh Valencia](#) , [Cornelia Mohren](#), [Anjali Ramakrishnan](#), [Marlene Merchert](#), [Jan C. Minx](#) & [Jan Christoph Steckel](#) 

[npj Climate Action](#) **3**, Article number: 74 (2024) | [Cite this article](#)

Article | [Open access](#) | Published: 16 May 2024

Systematic review and meta-analysis of ex-post evaluations on the effectiveness of carbon pricing

[Niklas Döbbeling-Hildebrandt](#) , [Klaas Miersch](#), [Tarun M. Khanna](#), [Marion Bachelet](#), [Stephan B. Bruns](#), [Max Callaghan](#), [Ottmar Edenhofer](#), [Christian Flachsland](#), [Piers M. Forster](#), [Matthias Kalkuhl](#), [Nicolas Koch](#), [William F. Lamb](#), [Nils Ohlendorf](#), [Jan Christoph Steckel](#) & [Jan C. Minx](#) 

[Nature Communications](#) **15**, Article number: 4147 (2024) | [Cite this article](#)

- Completely changes what is possible for evidence synthesis.
- Faster identification of what is relevant and why.
- Faster extraction of key information for synthesis and meta-analyses.
- Faster input for experts and policymakers.

AI for qualitative analysis → open-ended survey questions

> [Psychol Methods](#). 2025 Apr 7. doi: 10.1037/met0000753. Online ahead of print.

The use of large language models for qualitative research: The Deep Computational Text Analyser (DECOTA)

Lois Player¹, Ryan Hughes², Kaloyan Mitev¹, Lorraine Whitmarsh¹, Christina Demski¹,
Nicholas Nash¹, Trisevgeni Papakonstantinou³, Mark Wilson¹

Affiliations + expand

PMID: 40193412 DOI: [10.1037/met0000753](https://doi.org/10.1037/met0000753) 

JOURNAL ARTICLE

EDITOR'S CHOICE

Scaling Open-Ended Survey Responses Using LLM-Paired Comparisons

[Matthew R DiGiuseppe](#) , [Michael E Flynn](#)

Public Opinion Quarterly, Volume 90, Issue 3, Fall 2026, Pages 630–656,

<https://doi.org/10.1093/poq/nfag013>

Published: 27 March 2026

Current debates on AI, social science, and climate change research

ENVIRONMENTAL COST OF AI'S ENERGY USE

Carbon, Water and Land Footprints

GPT-4's training carbon footprint of 25,000 tonnes of CO₂e would require the sequestration capacity of 420,000 tree seedlings grown for 10 years, or about equal to the number of trees in 105 Hyde Parks in London.

The water footprint associated with training GPT-4 was about 600 million liters, enough to meet the minimum annual domestic water needs of 81,000 people in Sub-Saharan Africa, or to fill 237 Olympic-sized pools.

Training models like GPT-5 is estimated to have a carbon footprint of 42,000 tonnes of CO₂e, requiring 700,000 tree seedlings (about equal to the number of trees in 40 Central Parks in New York or 155 times the trees in Toronto's High Park over 10 years to offset).

The water footprint of GPT-5 training is estimated at 1 billion liters, enough to meet the annual domestic water needs of more than 135,000 people in Sub-Saharan Africa.

While training is highly resource-intensive, the continuous inference phase used to generate responses for billions of interactions is estimated to account for 80% to 90% of total AI energy use.

A typical ChatGPT-style text query is about 200 times more energy-intensive than text classification (such as spam filtering).

ENVIRONMENTAL COST OF AI'S ENERGY USE

Carbon, Water and Land Footprints

The global AI market is expanding rapidly, projected to grow from USD 189 billion in 2023 to nearly USD 5 trillion by 2033.

Generative AI accounted for over 20% of the global AI market in 2026, and is projected to reach 40% by 2030.

AI workloads accounted for roughly 20% of total data centers' electricity use in 2025 and are projected to grow to 40% by 2030.

Data centers' electricity use in 2025 had a carbon footprint of 189 million tonnes of CO₂e, which would require 3.2 billion tree seedlings grown over 10 years to offset, roughly the total number of trees in the entire United Kingdom.

The associated water footprint of projected 2030 electricity consumption of data centers is 9.3 trillion liters, or enough to meet the minimum annual domestic water needs of all 1.3 billion residents of Sub-Saharan Africa for a full year.

Projected global data centers' electricity consumption could exceed 945 TWh by 2030, accounting for almost 3% of projected global electricity use—enough to supply residential electricity to all 1.3 billion people in Sub-Saharan Africa for about 5.5 years.

AI boom means US is now 'investing more' in fossil-fuel power than China



Meta's \$13-billion Alberta data centre a potential boost for utilities and natural gas producers

ANDREW WILLIS >

Report warns AI data centre boom threatens Australia's energy transition

By national AI reporter Cam Wilson

AI

Tue 26 May

Revealed: landmark Scottish AI project has no prospect of meeting renewables promise

Exclusive: Government and developers privately
acknowledged Lanarkshire datacentre site had power
provision 'issue'

BREAKING | BUSINESS

Microsoft's AI Buildout Sent Carbon Emissions Up 25% Last Year

By [Ty Roush](#), Forbes Staff. Ty Roush is a breaking news reporter based in New ...

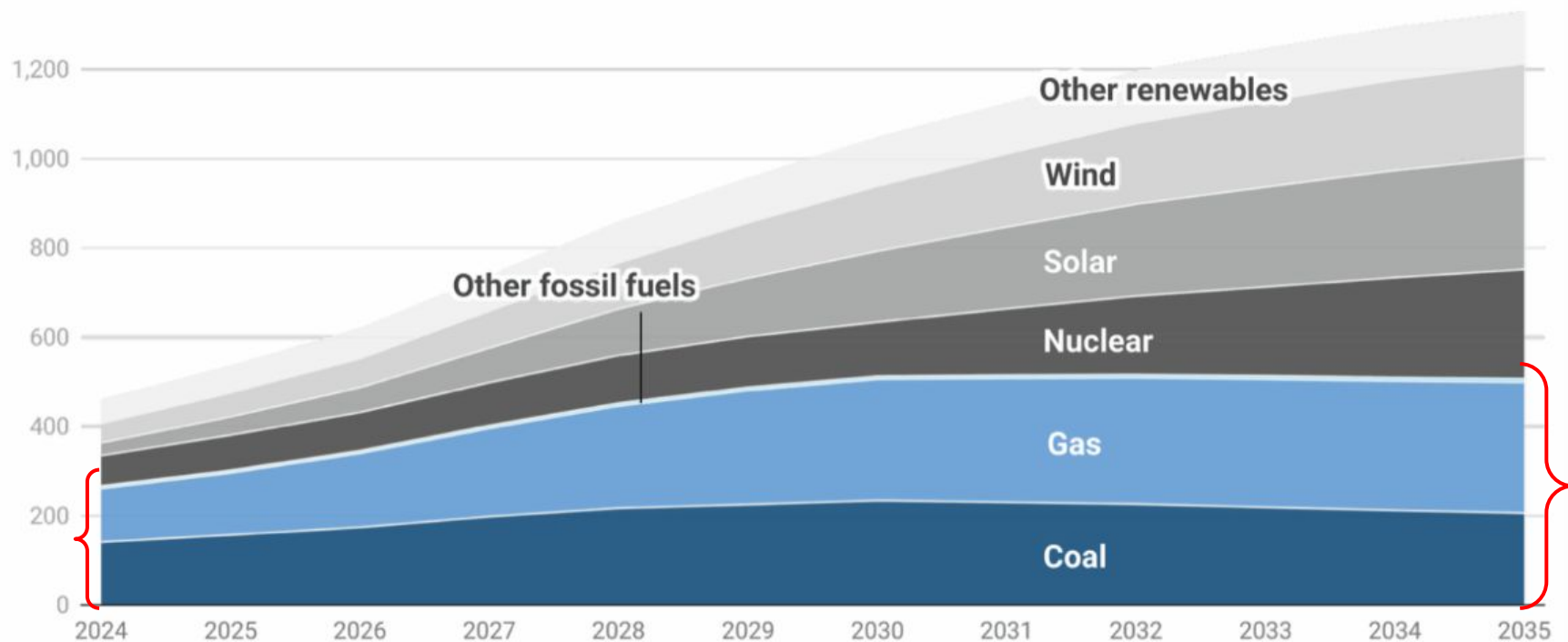


Follow Author

Published Jul 09, 2026, 02:45pm EDT

Fossil fuels currently provide the majority of power to data centres – but this is expected to shift over the next decade

Sources of electricity for data centres, TWh, 2024-2035



Source: IEA energy and AI report.

For more on these topics, be sure to attend Nikola's lecture on **12 August!**



The poster features a background image of a mountain landscape with a river. At the top center is the Climate Change AI logo, which consists of a globe and a network diagram. Below the logo is the text "Climate Change AI". The main title "Virtual Summer School 2026" is prominently displayed in the upper middle. A circular portrait of Nikola Milojevic-Dupont is centered below the title. To the right of the portrait, the text "Join to learn about:" is followed by "Sustainability Impacts of AI". At the bottom, the word "Partners:" is followed by logos for AfriClimate AI, data.org, CLIMATE INTERACTIVE, and cmcc.

Climate Change AI

Virtual Summer School 2026



Nikola Milojevic-Dupont
Core Team, Climate Change AI

Join to learn about:

Sustainability Impacts of AI

Partners:



AfriClimate AI
AI for a Climate Resilient Africa



data.org



CLIMATE
INTERACTIVE



cmcc
Centro Euro-Mediterraneo
per Cambiamenti Climatici

The background is a vibrant green with a complex, organic pattern of flowing, wavy lines. These lines vary in opacity, creating a sense of depth and movement, similar to smoke or liquid in motion. The overall effect is a dynamic and textured green field.

Question

16% of countries host AI-specialised cloud compute.
How much of that is concentrated across China and the US?

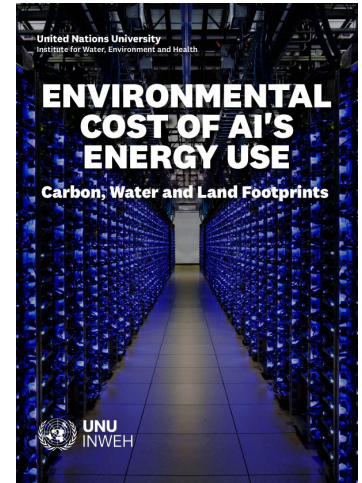
Estimate a value between 0 and 100%



<https://partici.fi/97320810>

16% of countries host AI-specialised cloud compute.
How much of that is concentrated across China and the US?

90%



Political implications

Significant geopolitical vulnerability

Who is behind the companies?

Accessibility to commercial models

Weakening of democracy

Security implications

MAXWELL ZEFF

BUSINESS FEB 12, 2026 2:00 PM

OpenAI's President Gave Millions to Trump. He Says It's

A.I. Is Starting to Wear Down Democracy

Content generated by artificial intelligence has become a factor in elections around the world. Most of it is bad, misleading voters and discrediting the democratic process.



Listen · 10:48 min



Share full article



136

Artificial intelligence

+ Add to myFT

OpenAI admits AI 'agent' caused major cyber breach by itself

AI lab's advanced models escaped testing 'sandbox' to hack Hugging Face

Validity crisis

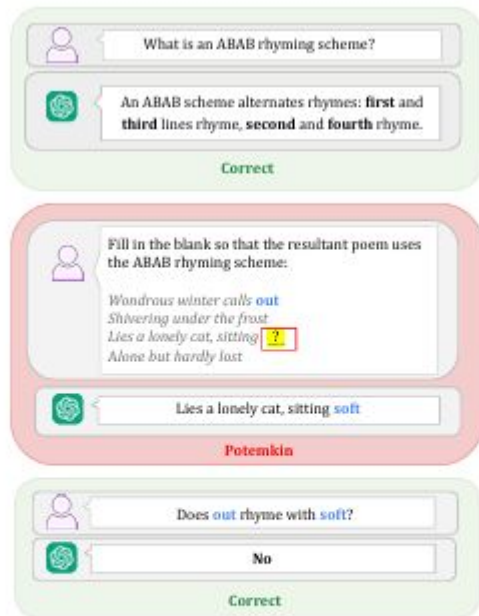


Figure 1. Illustration of potemkin understanding in a large language model. This example shows GPT-4o’s failure to apply its own conceptual explanation of an ABAB rhyming scheme.

Potemkin Understanding in Large Language Models

Marina Mancoridis¹ Keyon Vafa² Bec Weeks³ Sendhil Mullainathan¹

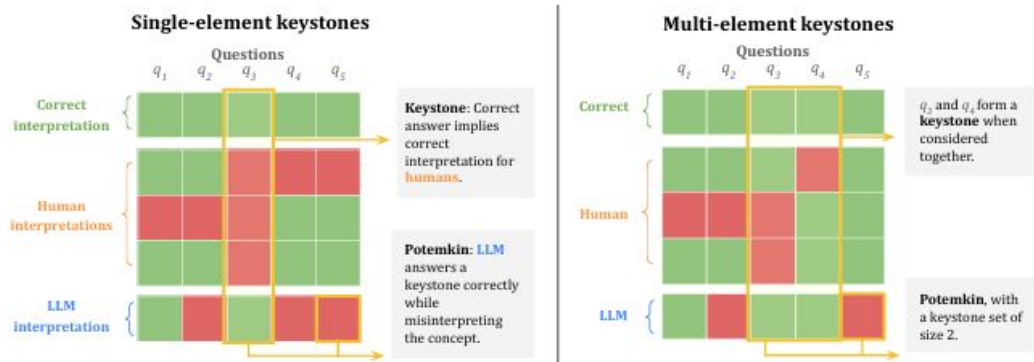


Figure 2. A schematic representation of keystones and potemkins. Rows represent interpretations of a concept and columns represent questions. Questions can either be interpreted **correctly** or **incorrectly**. A keystone is a set of questions that can only be interpreted correctly by a human who has understood the concept. An LLM has potemkin understanding when it correctly interprets all questions in a keystone but does not understand a concept.

Validity crisis

Potemkin Understanding in Large Language Models

Marina Mancoridis¹ Keyon Vafa² Bec Weeks³ Sendhil Mullainathan¹

Model	Potemkin rate, as measured by:		
	Classify	Generate	Edit
Llama-3.3	0.57 (0.06)	0.43 (0.09)	0.36 (0.05)
Claude-3.5	0.49 (0.05)	0.23 (0.08)	0.29 (0.04)
GPT-4o	0.53 (0.05)	0.38 (0.09)	0.35 (0.05)
Gemini-2.0	0.54 (0.05)	0.41 (0.09)	0.43 (0.05)
DeepSeek-V3	0.57 (0.05)	0.38 (0.09)	0.36 (0.05)
DeepSeek-R1	0.47 (0.05)	0.39 (0.09)	0.52 (0.05)
Qwen2-VL	0.66 (0.06)	0.62 (0.09)	0.52 (0.05)
Overall	0.55 (0.02)	0.40 (0.03)	0.40 (0.02)

Table 1. Potemkin rate on classify, generate, and edit tasks. The potemkin rate is the percent of questions about a concept that are answered incorrectly, conditional on a model defining that concept correctly. We scale the classification values so that chance-level performance corresponds with a potemkin rate of 1. Standard errors are in parentheses.

Validity crisis

Measuring what Matters: Construct Validity in Large Language Model Benchmarks

Andrew M. Bean^{1*} Ryan Othniel Kearns¹ Angelika Romanou²
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Jan Batzner^{3,4} Negar Foroutan² Chris Schmitz⁵ Karolina Korgul¹ Hunar Batra¹
Oishi Deb¹ Emma Beharry⁶ Cornelius Emde¹ Thomas Foster¹ Anna Gausen⁷
Maria Grandury^{8,9} Simeng Han¹⁰ Valentin Hofmann^{11,12} Lujain Ibrahim¹
Hazel Kim¹ Hannah Rose Kirk^{1,7} Fangru Lin¹
Gabrielle Kaili-May Liu¹⁰ Lennart Luetzgau⁷ Jabez Magomere¹ Jonathan Rystrom¹
Anna Sotnikova² Yushi Yang¹ Yilun Zhao¹⁰

Adel Bibi¹ Antoine Bosselut² Ronald Clark¹ Arman Cohan¹⁰ Jakob Foerster¹
Yarin Gal^{1,7} Scott A. Hale^{1,13} Inioluwa Deborah Raji¹⁴ Christopher Summerfield^{1,7}
Philip H.S. Torr¹ Cozmin Ududec⁷ Luc Rocher¹ Adam Mahdi^{1*}

¹University of Oxford ²EPFL ³Weizenbaum Institute Berlin

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⁶Stanford University ⁷UK AI Security Institute ⁸SomosNLP

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¹²University of Washington ¹³Meedan ¹⁴UC Berkeley

Abstract

Evaluating large language models (LLMs) is crucial for both assessing their capabilities and identifying safety or robustness issues prior to deployment. Reliably measuring abstract and complex phenomena such as ‘safety’ and ‘robustness’ requires strong *construct validity*, that is, having measures that represent what matters to the phenomenon. With a team of 29 expert reviewers, we conduct a systematic review of 445 LLM benchmarks from leading conferences in natural language processing and machine learning. Across the reviewed articles, we find patterns related to the measured phenomena, tasks, and scoring metrics which undermine the validity of the resulting claims. To address these shortcomings, we provide eight key recommendations and detailed actionable guidance to researchers and practitioners in developing LLM benchmarks.

Validity crisis

RESEARCH-ARTICLE | [OPEN ACCESS](#) | 

Gender Trouble in Language Models: An Empirical Audit Guided by Gender Performativity Theory

Authors:  [Franziska Sofia Hafner](#),  [Ana Valdivia](#),  [Luc Rocher](#) | [Authors Info & Claims](#)

FACCT '25: Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency • Pages 1677 - 1695
<https://doi.org/10.1145/3715275.3732112>

Published: 23 June 2025 [Publication History](#)

Analysis | [Open access](#) | Published: 12 December 2024

Generative language models exhibit social identity biases

[Tiancheng Hu](#) , [Yara Kyrychenko](#) , [Steve Rathje](#), [Nigel Collier](#), [Sander van der Linden](#) & [Jon Roozenbeek](#)

[Nature Computational Science](#) **5**, 65–75 (2025) | [Cite this article](#)

Bias in Large Language Models: Origin, Evaluation, and Mitigation

[Yufei Guo](#), [Muzhe Guo](#), [Juntao Su](#), [Zhou Yang](#), [Mengqiu Zhu](#), [Hongfei Li](#), [Mengyang Qiu](#), [Shuo Shuo Liu](#)

In research

- AI for survey responses?
 - ◆ Can LLMs reliably simulate human responses? *Current research tends towards **no**.*
- AI for qualitative interviews?
 - ◆ New methods for scaling up qualitative interviews which tend to be time consuming and expensive. *But what of the process might we lose through this?*
- Risks of deskilling?
 - ◆ AI changing the way we work, what we know, and what we know how to do...what could this mean for the development of research and education?
- Case of agents replacing research assistants – what will that do to early career researchers?
 - ◆ *Same applies to entry-level jobs.*

What do you think?

Can you think of an application where we shouldn't use AI?



<https://partici.fi/97320810>

The background of the slide is an abstract, fluid pattern of various shades of green. It consists of numerous overlapping, wavy lines and shapes that create a sense of movement and depth, resembling liquid or smoke. The colors range from a deep forest green to a bright, almost white-green at the edges of the waves.

What comes next?

Open questions

- Lots more to learn about how to mobilise people → *where can AI help?*
- Much more to understand about the pros and cons of how AI will impact various social and political challenges of the climate crisis.
- How we can shape the technology through the ways we use (or don't use) it?
- Climate change research, especially in the social sciences, is facing political headwinds...
How can keep the social sciences part of the conversation? On climate change, on AI, and on their intersection?

Action points!

- Carefully audit the strengths and weaknesses of AI tools to best be able to use them responsibly.
 - ◆ Treat it the way you would any other tool or method.
 - ◆ Keep the focus on the ‘nail’ (objective) not the ‘hammer’ (tool).
- Ask questions of AI applications.
 - ◆ What’s the training data? What potential biases do I need to know about? How can I mitigate them?
 - ◆ Does the cost justify the performance gain?
 - ◆ Are open source models available? Could they do the job?
- Resist the trend to apply LLMs without validation – establish norms of including validation strategies and practices in every application.
- Think (and speak!) creatively and critically about how to optimise human - AI teamwork. Keep the human in the loop!

The social sciences form a critical foundation on which to build climate action towards decarbonisation.

- Diagnose barriers and identify solutions for establishing crucial social and political support across stakeholders → necessary for the uptake of technological advances.
- AI can accelerate parts of this process, but it is best used **wisely not widely**.

The background is a vibrant green with a complex, organic pattern of flowing, wavy lines. These lines vary in opacity, creating a sense of depth and movement, similar to smoke or liquid in motion. The overall effect is a dynamic and textured green field.

Thank you for your attention.